

## Employment

### *Software Engineer, Compiler*

May 2025 – Present

RelationalAI, Logic Engine team

Remote

- Design and implement the procedural extension to RelationalAI’s query language
  - Scale adoption from internal algorithms to several enterprise workloads
  - Optimize previously intractable queries to sub-hour runtimes
- Build the GQL-style graph pattern matching extension for the query language
- Maintain the database engine’s Protocol Buffer-based entry point (~100k queries/day)
- Develop an event-based user-facing logging tool to diagnose slow queries

### *Graduate Researcher*

Sep 2022 – Apr 2025

Northeastern University, Programming Research Lab

Boston, USA

- Designed a language to express optimization problems under uncertainty [1]
  - Implemented a novel binary decision diagram algorithm to compute optimal values
  - Outperformed a state-of-the-art probabilistic programming language by up to 10x
- Co-led the first empirical study of online proof assistant communities [2]
  - Applied established developer lifecycle models to characterize contribution patterns

### *R&D Intern*

May 2024 – Dec 2024

RelationalAI, Logic Engine team

Remote

- Shipped a decompiler that lifts optimized Datalog back to source query
- Integrated it into the production query profiler to automatically flag performance bottlenecks

## Education

### *Master of Science, Computer Science*

Sep 2022 - May 2025

Northeastern University

Boston, USA

### *Bachelor of Science, Mathematics and Philosophy*

Aug 2018 - May 2022

Carnegie Mellon University

Pittsburgh, USA

## Skills

**Programming languages:** Julia, Python, Datalog, OCaml, Rust, Go, C/C++, Haskell, Standard ML

**Tools:** Unix, Git, Jira, Docker, Datadog, Observe, Snowflake, LLVM/MLIR

## Awards and Recognition

First Place, ACM Student Research Competition at PLDI (top PL conference) **2023**

Scholarship Recipient, Programming Languages Mentoring Workshop (PLMW) at PLDI **2023**

University and College Honors, Carnegie Mellon University **2022**

## Publications

- [1] Minsung Cho, John Gouwar, and Steven Holtzen. “Scaling Optimization over Uncertainty via Compilation”. In: *Proc. ACM Program. Lang.* 9.OOPSLA1 (2025). DOI: 10.1145/3720500.
- [2] Gwennyth Lincroft\*, Minsung Cho\*, et al. “Thirty-Three Years of Mathematicians and Software Engineers: A Case Study of Domain Expertise and Participation in Proof Assistant Ecosystems”. In: *21st IEEE/ACM International Conference on Mining Software Repositories, MSR 2024*. ACM, 2024, pp. 1–13. DOI: 10.1145/3643991.3644908. \*Joint first authors.
- [3] Minsung Cho, Seth Hoisington, Roger Nichols, and Brian Udall. “The Krein-von Neumann extension of a regular even order quasi-differential operator”. In: *Opuscula Mathematica* 41.6 (2021), pp. 805–841. DOI: 10.7494/OpMath.2021.41.6.805.